

Joseph Wallace

Computational Chemist | Ligand Prioritisation
Structure-Based Modelling

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Professional Summary

Computational chemist focused on structure-guided molecular modelling, ligand prioritisation, and physically grounded analysis across biomolecular systems. Currently building reproducible ligand-screening, docking, rescoring, and scientific Python workflows that support the cycle from screening to binding-mode inspection, rationalisation, and prioritisation. Strong focus on benchmark-aware evaluation and interpretable computational outputs for discovery-relevant decisions.

Core Skills

Molecular Modelling & CADD: Structure-based design, ligand-based analysis, virtual screening, docking/rescoring workflows, molecular interaction analysis, binding-mode interpretation, comparative ranking and benchmark evaluation.

Scientific Computing & Cheminformatics: Python, NumPy, pandas, SciPy, matplotlib, RDKit, reproducible analysis pipelines, workflow automation, data visualisation.

Simulation Methods: Molecular dynamics, enhanced sampling, umbrella sampling, metadynamics, Hamiltonian replica exchange, free-energy-oriented analysis.

Molecular Modelling Tools: PyMOL, VMD, AMBER, GROMACS, PLUMED, Linux/HPC environments.

Additional familiarity: Schrödinger suite, Maestro, LigPrep, Epik.

Professional Experience

Dec 2022–Present **Postdoctoral Researcher**, *Italian Institute of Technology (IIT) / University of Padova*, Italy

- Based in Marco De Vivo's Molecular Modeling and Drug Discovery lab at IIT Genoa, with initial University of Padova appointment followed by IIT and Sustainable Technologies flagship support.
- Lead computational modelling and interaction analysis on nanoparticle–molecule systems, with emphasis on physically grounded simulation, molecular recognition, and reproducible scientific workflows.
- Designed and maintained Python-based pipelines for automated processing, visualisation, and benchmark-style comparison across systems, producing interpretable and decision-ready outputs.
- Applied molecular simulation and enhanced-sampling methods to study structure, dynamics, and binding behaviour in complex molecular systems, using the results to support prioritisation, hypothesis generation, and design-oriented iteration.

2021–2022 **PhD Research Attachment**, *Université Sorbonne Paris Nord*, France

- Supported research continuity during laboratory relocation; contributed to computational research within an international academic environment.

2019–2021 **PhD Research Attachment**, *A*STAR Institute of High Performance Computing*, Singapore

- Conducted molecular simulation, structural analysis, and computational biochemistry research in an HPC environment, with emphasis on scalable workflows, reproducible analysis, and physically interpretable results.

Education

2017–2022 **PhD, Computational Biochemistry**, *University of Liverpool*, UK

Thesis: How does constraining folded peptides within dense self-assembled monolayers on gold nanoparticles affect their structure? Combined molecular simulation and experimental data to study structure, dynamics, and interactions in peptide–nanoparticle systems.

2014–2017 **BSc Biochemistry, First Class Honours**, *University of Liverpool*, UK

Selected Projects

Docking + Rescoring Benchmark for Ligand Ranking

- Built a reproducible ligand-prioritisation workflow to test whether rescoring improves ranking beyond a docking-only baseline on a curated protein–ligand benchmark.
- Methods emphasis: docking baseline, rescoring, RDKit/scientific Python analysis, benchmark evaluation, and binding-mode/interaction review.
- Focused on virtual screening logic, ranking analysis, benchmark design, and explicit failure-mode evaluation under realistic screening constraints rather than inflated performance claims.
- Relevance: designed to support interpretable ligand prioritisation and design-oriented decision-making.
- *Links:* Project page | Repository

Simulation Analysis Pipelines for Molecular Systems

- Designed Python-based workflows for processing, comparing, and visualising molecular simulation outputs across multiple systems.
- Emphasis on reproducibility, interaction-focused analysis, and workflow reuse for design-support and research decision-making.
- Relevance: turns simulation outputs into comparable, inspectable evidence for hypothesis generation and prioritisation.

Selected Publications

- 1 Wallace, J., Riccardi, L., Mancin, F., & De Vivo, M. (2025). A Scoring Function for Monolayer-Protected Gold Nanoparticles Capable of Recognizing Small Organic Molecules in Solution. *Journal of Chemical Theory and Computation*, **21**(21), 11070–11078. DOI: 10.1021/acs.jctc.5c01278
- 2 Franco-Ulloa, S., Cesari, A., Zanoni, G., Riccardi, L., Wallace, J., et al. (2025). Rational design of gold nanoparticle-based chemosensors for detection of the tumor marker 3-methoxytyramine. *Chemical Science*, **16**, 6282–6289. DOI: 10.1039/D4SC08758E
- 3 Wallace, J. (2022). *How Does Constraining Folded Peptides within Dense Self-Assembled Monolayers on Gold Nanoparticles Affect Their Structure? An Experimental and Computational Investigation*. PhD thesis, University of Liverpool.

Earlier Structural Biology Background

BSc project	Solving the crystal structure of Cytochrome bc ₁ for the development of antimalarials.
Research context	Earlier training connected protein structure, biochemistry, and experimental evidence with later computational modelling and structure-focused analysis.